

CivicPulse

Product Specification / Spec Sheet

1. Overview

CivicPulse is a responsive civic-issue reporting web application for residents and local authorities. Citizens can report issues such as potholes, garbage, broken streetlights, water leaks, drainage problems, and unsafe infrastructure, including photos and location details. Authorities can review reports, assign them to departments, update statuses, and monitor city-wide issues through dashboards and maps.

2. Goals

- Make reporting local civic problems simple and accessible.
- Help authorities organize, prioritize, assign, and track reported issues.
- Provide clear visibility into the lifecycle of each civic complaint.
- Use AI-assisted suggestions for issue category, severity, and responsible department.
- Provide a responsive experience across desktop and mobile devices.

3. Core User Roles

Role	Main capabilities
Citizen / Resident	Create civic issue reports, attach photos, provide location/details, and track report status.
Authority / Administrator	View reports, inspect issue details, use dashboards/maps, assign departments, and update statuses.

4. Main Issue Categories

- Potholes and damaged roads
- Garbage / waste management
- Streetlight and public-lighting problems
- Water leaks and water-supply issues
- Drainage and sewage problems
- Unsafe or damaged public infrastructure

5. Key Features

Issue Reporting: Residents can submit an issue with category, description, photo(s), and location information.

AI Assistance: Suggests a likely issue category, severity level, and responsible department.

Issue Tracking: Residents can view the progress of submitted issues through a status lifecycle.

Authority Dashboard: Authorities can view city-wide reports and high-level issue statistics.

Map View: Reports can be viewed geographically to help identify local clusters and priorities.

Assignment: Authorities can assign reports to the appropriate department.

Status Updates: Authorities can move issues through defined workflow states.

Responsive UI: The interface reflows for phones and wider desktop screens.

6. Application Pages / Navigation

- Home / Overview
- Report an Issue
- My Reports / Issue Tracking
- Issue Details
- Authority Dashboard
- Reports / Map View
- Department / Assignment Management
- Profile / Settings

7. UI / Look & Feel

- Clean civic-tech visual style with clear cards, panels, forms, and status indicators.
- Primary visual direction uses deep civic blue (#0F4C81), bright teal accents, neutral greys, and white.
- Top header contains the CivicPulse identity and global navigation.
- Authority views use a collapsible left-side navigation on wider screens.
- Key pages can use a contextual right-hand panel on desktop.
- Dark-mode support is part of the visual design direction.

8. Data & Demo Behavior

- The first implementation uses demo/sample data for the core experience.
- Report records contain issue information such as category, description, location, severity, department, status, timestamps, and media references.
- Session persistence is used where needed for the demo workflow.
- The deployed application uses an application database generated from the product specification.

9. Issue Status Lifecycle

Stage	Purpose
Submitted	Issue has been reported by a citizen.
Under Review	Authority is reviewing the report.
Assigned	The issue has been assigned to a responsible department/team.
In Progress	Work to resolve the issue is underway.
Resolved	The issue has been addressed.

10. Validation & Interaction Requirements

- Required fields should be clearly identified.
- Forms should provide understandable validation and error states.

- Loading, empty, success, and failure states should be represented where applicable.
- Actions that change report status or assignment should provide clear feedback.
- Navigation should remain understandable on both desktop and mobile layouts.

11. Accessibility & Responsiveness

- Readable typography and sufficient visual contrast.
- Controls should be usable with keyboard navigation where applicable.
- Images and icons should have meaningful labels or alternative text when needed.
- Layouts should reflow into a single-column mobile experience.

12. First Release Scope

The first release focuses on the core CivicPulse UI/navigation, reporting workflow, issue tracking, authority management views, demo data, and responsive behavior. External APIs and production authentication are not required for the initial demonstration; AI assistance may use a client-side/mock implementation for the first release.

13. Acceptance Criteria

- A resident can navigate to the reporting flow and submit a civic issue using the available fields.
- A submitted report can be viewed and tracked through its status lifecycle.
- An authority user can review reports and see useful issue information.
- An authority user can assign an issue to a department and update its status.
- Dashboard/map views communicate the overall issue situation using the available demo data.
- The application remains usable on both desktop and mobile screen sizes.